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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : IT-801A

ADVANCED COMPUTER ARCHITECTURE

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A
(Multiple Choice Type Questions)

1. Choose the correct alternatives for the following : $10 \times 1 = 10$

i) What will be the speed up for a 4 segment linear pipeline when the number of instruction $n = 64$?
   a) 4·5 b) 3·82
   c) 8·16 d) 2·95.

ii) Virtual address can be divided into some fixed size fields of
   a) segments b) blocks
   c) pages d) tags.

iii) Which is not the property of a memory module ?
   a) Inclusion b) Capability
   c) Locality d) Consistency.

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iv) In absence of TLB, to access a physical memory location in a page-memory system how many memory accesses are required ?
   a) 1 b) 2
   c) 3 d) 4.

v) A pipeline stage
   a) is sequential circuit
   b) is combinational circuit
   c) both (a) and (b)
   d) none of these.

vi) Which of the following is not the possible cause of data hazard ?
   a) RAW b) RAR
   c) WAW d) WAR.

vii) The time to access shared memory is same in which of the following shared memory multiprocessor models ?
   a) UMA b) COMA
   c) SIMD d) NUMA.

viii) Utilization pattern of successive stages of a synchronous pipeline can be specified by
   a) Reservation table b) Truth table
   c) Excitation table d) Periodic table.

ix) Array processing is possible in which type of processor ?
   a) SIMB b) MIMD
   c) MISD d) SISD.

x) Conflict miss is occurred in which type of memory mapping ?
   a) Set associative b) Direct
   c) Associative d) All of these.

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GROUP – B
(Short Answer Type Questions)

Answer any *three* of the following $3 \times 5 = 15$

2. Compare, Superscalar, Superpipeline and superscalar superpipelined processor.
3. Explain Flynn's classification of computers with suitable diagrams.
4. Explain the main factors influencing the performance of interconnection networks.
5. Assume a main memory of size 32K × 12, Cache memory of size 512 × 12 and block size of 1 word. For direct mapping what would be the size of the tag and index fields ? Explain Data Hazard with example. 3 + 2
6. Explain *S* access and *C* access memory organization.

GROUP – C
(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. Consider the following reservation table :

	1	2	3	4
S1	X			X
S2		X		
S3			X	

Write down the forbidden latencies and initial collision vector. Draw the state diagram for scheduling the Pipeline. Find out the simple and greedy cycle and MAL. If the pipeline clock rate is 25 MHz then what is the throughput of the pipeline ? What are the bonds on MAL ? 3 + 3 + 3 + 3 + 3

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8. a) Differentiate between multiprocessors and multicomputers based on their resource sharing, structure and processor communication.
   b) Explain 10 subsystems in lightly coupled multiprocessor system with neat sketch. $7\frac{1}{2} + 7\frac{1}{2}$
9. a) How does cache memory effect the throughput of a computer system ?
   b) Distinguish between Write back and Write through cache.
   c) What is the effect of memory bandwidth over effective memory access time ?
   d) What is cache coherence ? How can we overcome this problem ? 3 + 5 + 4 + 3
10. a) Compare the advantages and disadvantages of Direct mapping and Associative mapping.
    b) A system with four way set associative cache has 128 blocks in the cache memory, each block contains 16 words (8 bit words) and the main memory has 16384 words. How many blocks are there in the main memory and what is the size of tag field ?
    c) Explain locality of reference. 5 + 5 + 5
11. Write short notes on any *three* of the following : $3 \times 5 = 15$
    a) Omega Network
    b) Array processor
    c) Reservation table
    d) Multiport network
    e) Power PC.

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